

Ilyen-és-ilyen alakú integrálok (helyettesítéses integrál), parciális integrálás

1. Ilyen-és-ilyen alakú integrálok (helyettesítéses integrál)

$$\boxed{\int g^n(x)g'(x)dx = \frac{g^{n+1}(x)}{n+1} + C} \quad (n \neq -1)$$

$$\begin{aligned} \text{(a)} \quad & \int \sin^7(x) \cdot \cos x \, dx, & \text{(b)} \quad & \int \frac{1}{x \ln^3 x} \, dx, & \text{(c)} \quad & \int \sqrt[3]{e^x + x} \cdot (e^x + 1) \, dx, \\ \text{(d)} \quad & \int (x^2 + 2x + 3)^5 \cdot (x + 1) \, dx, & \text{(e)} \quad & \int \frac{x^3 + 2}{(x^4 + 8x - 1)^2} \, dx, & \text{(f)} \quad & \int \frac{e^{4x} + 1}{\sqrt[4]{e^{4x} + 4x}} \, dx. \end{aligned}$$

$$\boxed{\int \frac{g'(x)}{g(x)} dx = \ln g(x) + C}$$

$$\text{(g)} \quad \int \tan x \, dx, \quad \text{(h)} \quad \int \frac{1}{x \ln x} \, dx, \quad \text{(i)} \quad \int \frac{1}{(x^2 + 1) \arctan x} \, dx.$$

$$\boxed{\int f(g(x))g'(x)dx = F(g(x)) + C} \quad (F' = f)$$

$$\begin{aligned} \text{(k)} \quad & \int \frac{\cos x}{\cos^2(\sin x)} \, dx, & \text{(l)} \quad & \int \frac{\ln x}{x} \, dx, & \text{(m)} \quad & \int \frac{e^x + e^{2x}}{\cos^2(2e^x + e^{2x})} \, dx, \\ \text{(o)} \quad & \int e^{x+\ln x} \cdot \frac{1+x}{x} \, dx, & \text{(p)} \quad & \int \frac{e^{x+1}}{e^{2x+2} + 1} \, dx, & \text{(q)} \quad & \int \sinh(\sin 2x + \cos 2x)(\sin 2x - \cos 2x) \, dx. \end{aligned}$$

2. Parciális integrálás

$$\boxed{\int Fg = FG - \int fG} \quad (F' = f, G' = g)$$

$$\begin{aligned} \text{(a)} \quad & \int (x+1) \sin x \, dx, & \text{(b)} \quad & \int x^2 \cos 2x \, dx, & \text{(c)} \quad & \int (2x+1)e^{x+3} \, dx \\ \text{(d)} \quad & \int \frac{\ln x}{x} \, dx, & \text{(e)} \quad & \int \frac{\arctan x}{x^2 + 1} \, dx, & \text{(f)} \quad & \int \ln x \, dx \\ \text{(g)} \quad & \int e^x \cos x \, dx, & \text{(h)} \quad & \int e^{2x-1} \sin(3x+4) \, dx, & \text{(i)} \quad & \int x^2 \arctan x \, dx \end{aligned}$$

3. Integrálás a helyettesítés elvégzésével

$$\int f(g(x))g'(x) \, dx = \int f(u) \, du \Big|_{u=g(x)} = F(u)|_{u=g(x)} + C$$

$$\text{(a)} \quad [u = e^x] \quad \int \frac{e^x}{1+e^{2x}} \, dx, \quad \text{(b)} \quad [u = \sqrt{x}] \quad \int \frac{2x}{1+\sqrt{x}} \, dx, \quad \text{(c)} \quad [x = \cos t] \quad \int \frac{1}{\sqrt{1-x^2}} \, dx.$$